

SALES ANALYSIS DASHBOARD

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Course: Business Analysis

Batch:164

1. Introduction

In today's competitive business environment, organizations rely on data-driven decision-making to improve sales performance, profitability, and customer satisfaction. This project focuses on analyzing a retail sales dataset using Microsoft Power BI to uncover valuable business insights.

The objective of this project is to develop an interactive Sales Analysis Dashboard that enables stakeholders to monitor sales performance, analyze customer behavior, evaluate product profitability, and identify key business trends. The dashboard provides visual insights that support strategic decision-making and business growth.

2. Project Objectives

The primary objectives of this project are:

- Analyze overall sales and profit performance.
- Identify top-performing products and customers.
- Evaluate sales distribution across regions and customer segments.
- Study the relationship between discounts and profitability.
- Identify loss-making products and regions.
- Create interactive dashboards for business users.
- Provide advanced analytical insights through decomposition trees and key influencer analysis.

3. Dataset Overview

The dataset contains sales transaction records collected from a retail business.

Key Attributes

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Region
- State
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Ship Mode

The dataset consists of approximately 9,994 records and multiple business dimensions related to customers, products, and sales transactions.

4. Data Preparation

Data preparation was performed using Power Query in Power BI.

Data Cleaning Activities

- Verified data types for all columns.
- Converted Order Date and Ship Date into date format.
- Checked for missing values and data inconsistencies.
- Created Shipping Days column using the difference between Ship Date and Order Date.
- Established relationships between Sales and People tables.
- Created DAX measures for business analysis.

DAX Measures Created

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Total Quantity

- Profit Margin
- Average Discount
- Average Shipping Days
- Loss Orders
- Average Order Value

5. Dashboard Development

The Power BI solution was divided into four dashboard pages.

Page 1: Executive Dashboard

This dashboard provides a high-level overview of business performance.

Visuals Included:

- Total Sales KPI
- Total Profit KPI
- Total Orders KPI
- Total Customers KPI
- Total Quantity KPI
- Profit Margin KPI
- Sales Trend Analysis
- Sales by Category
- Profit by Category
- Sales by Region
- Top 10 Products

Purpose:

To provide management with a quick snapshot of business performance.

Page 2: Customer and Product Analysis

This dashboard focuses on customer behavior and product performance.

Visuals Included:

- Top Customers by Sales
- Sales by Customer Segment
- Average Order Value
- Sales by Category
- Profit by Sub-Category

- Top Products by Profit
- Bottom Products by Profit
- Quantity Sold Analysis
- Customers by Region

Purpose:

To identify valuable customers and profitable products.

Page 3: Profitability and Discount Analysis

This dashboard evaluates profitability and discount impacts.

Visuals Included:

- Total Profit KPI
- Profit Margin KPI
- Average Discount KPI
- Loss Orders KPI
- Discount vs Profit Scatter Plot
- Profit by Region
- Profit by Segment
- Profit by Shipping Mode
- Bottom 10 States by Profit
- Bottom 10 Products by Profit

Purpose:

To understand the factors affecting profitability and identify areas requiring corrective action.

Page 4: Forecasting and Advanced Insights

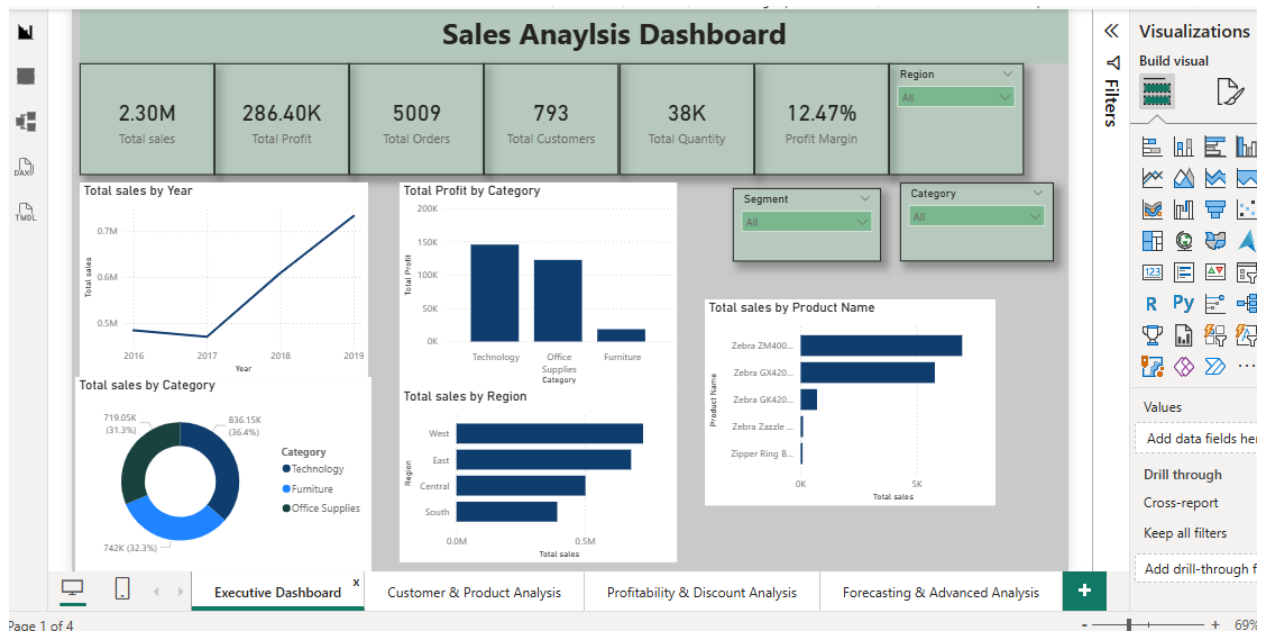
This dashboard provides advanced analytical capabilities.

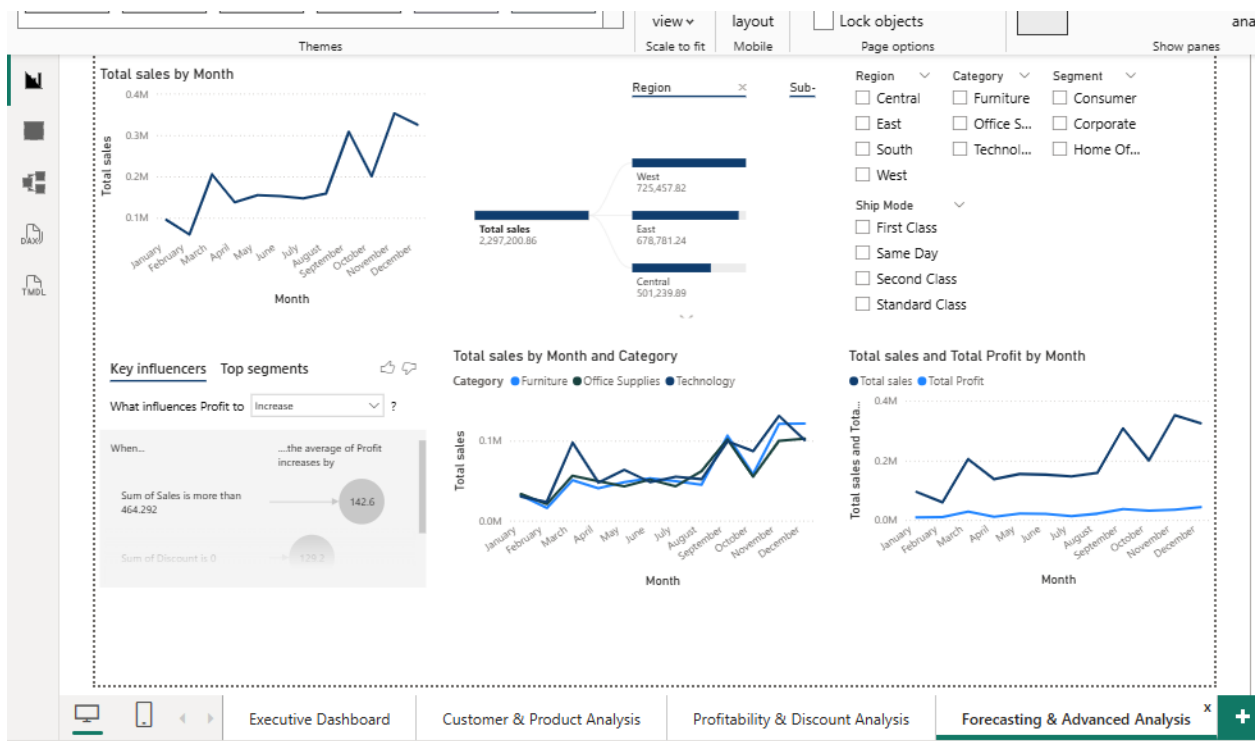
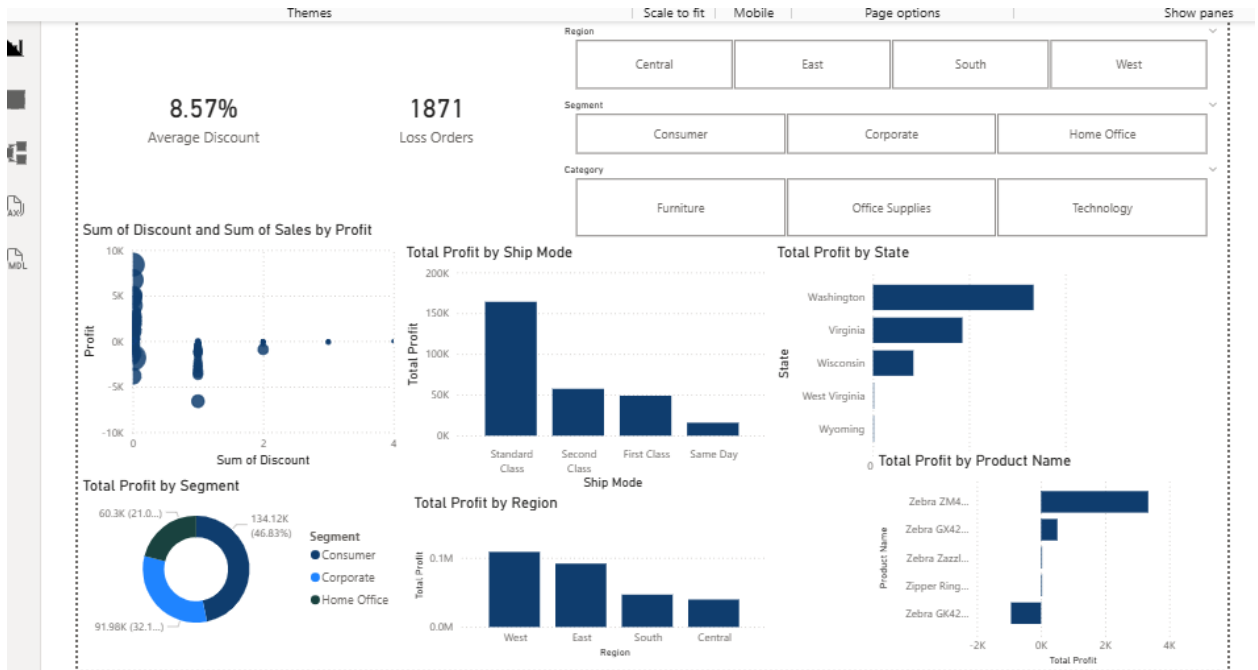
Visuals Included:

- Monthly Sales Trend
- Sales vs Profit Trend
- Category Trend Analysis
- Decomposition Tree
- Key Influencers Visual

Purpose:

To identify sales drivers and support strategic planning.





6. Key Findings

The analysis revealed several important business insights:

1. Technology products generated the highest sales revenue.
2. Certain customer segments contributed significantly to total sales.
3. High discounts often reduced profitability.
4. Some products generated consistent losses despite strong sales.
5. Regional performance varied significantly across the business.
6. A small group of customers contributed a large share of total revenue.
7. Profitability was concentrated within specific product categories and regions.
8. Shipping mode influenced profit performance.
9. Several states showed negative profitability and require business attention.
10. Sales trends indicated opportunities for future growth.

7. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Reduce excessive discounting on low-margin products.
- Focus marketing efforts on high-value customer segments.
- Promote top-performing products and categories.
- Review pricing strategies for loss-making products.
- Improve inventory planning for high-demand categories.
- Investigate causes of low profitability in underperforming regions.
- Enhance customer retention programs for top customers.
- Use advanced analytics regularly to monitor business performance.

8. Conclusion

This project successfully demonstrates the use of Microsoft Power BI for business intelligence and sales analytics. The interactive dashboard provides valuable insights into sales performance, customer behavior, product profitability, and business trends.

The findings can assist management in making informed decisions, improving operational efficiency, and increasing overall profitability. The project highlights the effectiveness of data visualization and analytics in transforming raw business data into actionable insights.

Tools Used

- Microsoft Power BI

- Power Query
- DAX (Data Analysis Expressions)
- Microsoft Excel

Thankyou